Управление логическими томами (LVM)

Эйвази Мани

13 марта 2026

Contents

Цели и задачи работы **2**
Цель лабораторной работы . . . 2

Процесс выполнения лабораторной работы **3**
Создание физического тома . . . 3
Проверка VG . . . 4
Создание файловой системы . . . 5
Редактирование fstab . . . 6
Проверка монтирования . . . 7
Добавление нового раздела . . . 8
Увеличение VG . . . 9
Увеличение размера LV . . . 10
Уменьшение размера LV . . . 11
Создание PV/VG/LV . . . 12
Настройка fstab . . . 13

Создание нового PV для расширения 14

Выводы по проделанной работе **15**
Вывод 15

Цели и задачи работы

Цель лабораторной работы

Получить навыки создания, управления и модификации логических томов LVM.

Процесс выполнения лабораторной работы

Создание физического тома

-

```
root@localhost:/home/manieyvazi# pvcreate /dev/sdb1
  Physical volume "/dev/sdb1" successfully created.
```

Рис. 1 — Очистка и разметка диска /dev/sdb

Проверка VG

-.

```
root@localhost:/home/manieyvazi# pvcreate /dev/sdc
  Physical volume "/dev/sdc" successfully created.
root@localhost:/home/manieyvazi# vgcreate -s 4M vg /dev/sdb1 /dev/sdc
  Volume group "vg" successfully created
root@localhost:/home/manieyvazi# vgdisplay vg
  --- Volume group ---
  VG Name               vg
  System ID
  Format                lvm2
  Metadata Areas        2
  Metadata Sequence No  1
  VG Access             read/write
  VG Status             resizable
  MAX LV                0
  Cur LV                0
  Open LV               0
  Max PV                0
  Cur PV                2
  Act PV                2
  VG Size               1016.00 MiB
  PE Size               4.00 MiB
  Total PE              254
  Alloc PE / Size       0 / 0
  Free  PE / Size       254 / 1016.00 MiB
  VG UUID               ctQKbv-BHsn-hMzX-qSTi-0V7C-K2xB-wnHCPE
```

Рис. 2 — Вывод vgs

Создание файловой системы

•

```
root@localhost:/home/manieyvazi# lvcreate -n lv1 -L 20G vg
  Volume group "vg" has insufficient free space (254 extents): 5120 required.
root@localhost:/home/manieyvazi# lvcreate -n lv1 -L 10G vg
  Volume group "vg" has insufficient free space (254 extents): 2560 required.
root@localhost:/home/manieyvazi# lvcreate -n lv1 -L 2G vg
  Volume group "vg" has insufficient free space (254 extents): 512 required.
root@localhost:/home/manieyvazi# lvcreate -n lv1 -L 200M vg
  Logical volume "lv1" created.
root@localhost:/home/manieyvazi# lvcreate -n lv@ -L 300M vg
  Logical volume name "lv@" is invalid.
  Run `lvcreate --help' for more information.
root@localhost:/home/manieyvazi# lvcreate -n lv2 -L 300M vg
  Logical volume "lv2" created.
root@localhost:/home/manieyvazi# mkfs.ext4 /dev/vg/lv1
mke2fs 1.47.1 (20-May-2024)
Creating filesystem with 204800 1k blocks and 51200 inodes
Filesystem UUID: 79c0fd0a-b491-4773-8e10-cf4844219ffd
Superblock backups stored on blocks:
        8193, 24577, 40961, 57345, 73729
```

Рис. 3 — Создание файловой системы

Редактирование fstab

-.

```
#
# /etc/fstab
# Created by anaconda on Tue Nov 18 11:38:02 2025
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=8a031512-0250-4e89-af86-96ba8fe9a665 /                       xfs     defaults        0 0
UUID=ee219ed5-dbf2-4e2d-8830-766d91c81b30 /boot                   xfs     defaults        0 0
UUID=f1486dd0-368a-47c6-bafc-2cd97baea386 none                    swap    defaults        0 0
```

Рис. 4 — Редактирование fstab

Проверка монтирования

-.

```
root@localhost:/home/manieyvazi# lvcreate -n lv1 -L 20G vg
  Volume group "vg" has insufficient free space (254 extents): 5120 required.
root@localhost:/home/manieyvazi# lvcreate -n lv1 -L 10G vg
  Volume group "vg" has insufficient free space (254 extents): 2560 required.
root@localhost:/home/manieyvazi# lvcreate -n lv1 -L 2G vg
  Volume group "vg" has insufficient free space (254 extents): 512 required.
root@localhost:/home/manieyvazi# lvcreate -n lv1 -L 200M vg
  Logical volume "lv1" created.
root@localhost:/home/manieyvazi# lvcreate -n lv@ -L 300M vg
  Logical volume name "lv@" is invalid.
  Run `lvcreate --help' for more information.
root@localhost:/home/manieyvazi# lvcreate -n lv2 -L 300M vg
  Logical volume "lv2" created.
root@localhost:/home/manieyvazi# mkfs.ext4 /dev/vg/lv1
mke2fs 1.47.1 (20-May-2024)
Creating filesystem with 204800 1k blocks and 51200 inodes
Filesystem UUID: 79c0fd0a-b491-4773-8e10-cf4844219ffd
Superblock backups stored on blocks:
        8193, 24577, 40961, 57345, 73729
```

Рис. 5 — Монтирование LV

Добавление нового раздела

-.

```
Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

This disk is currently in use - repartitioning is probably a bad idea.
It's recommended to umount all file systems, and swapoff all swap
partitions on this disk.


Command (m for help): p

Disk /dev/sdb: 512.38 MiB, 537269248 bytes, 1049354 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xc9cf3cef

Device     Boot Start     End Sectors   Size Id Type
/dev/sdb1        2048 1049353 1047306 511.4M fd Linux raid autodetect

Command (m for help): o
Created a new DOS (MBR) disklabel with disk identifier 0xcaf04faa.

Command (m for help): p
Disk /dev/sdb: 512.38 MiB, 537269248 bytes, 1049354 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcaf04faa

Command (m for help): w
The partition table has been altered.
```

Рис. 6 — Создание /dev/sdb2

Увеличение VG

-.

```
root@localhost:/home/manieyvazi# fdisk /dev/sdc

Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0xab06fbc4.

Command (m for help): p
Disk /dev/sdc: 512.24 MiB, 537122304 bytes, 1049067 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xab06fbc4

Command (m for help): o
Created a new DOS (MBR) disklabel with disk identifier 0xfb8266d9.

Command (m for help): p
Disk /dev/sdc: 512.24 MiB, 537122304 bytes, 1049067 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xfb8266d9

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
```

Рис. 7 — Увеличение VG

Увеличение размера LV

-.

```
root@localhost:/home/manieyvazi# partprobe /dev/sdb
Error: Partition(s) 1 on /dev/sdb have been written, but we have been unable to inform the kernel of the change, prob
ably because it/they are in use.  As a result, the old partition(s) will remain in use.  You should reboot now before
 making further changes.
root@localhost:/home/manieyvazi# cat /proc/partitions
major minor  #blocks  name

   8       16     524677 sdb
   8       17     523653 sdb1
   8        0   41943040 sda
   8        1       1024 sda1
   8        2    1048576 sda2
   8        3   40891392 sda3
   8       32     524533 sdc
   8       48     524554 sdd
  11        0      51908 sr0
 253        0   36696064 dm-0
 253        1    4194304 dm-1
 253        2     204800 dm-2
 253        3     307200 dm-3
root@localhost:/home/manieyvazi# fdisk --list /dev/sdb
Disk /dev/sdb: 512.38 MiB, 537269248 bytes, 1049354 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xcaf04faa
root@localhost:/home/manieyvazi#
```

Рис. 8 — Увеличение LV

Уменьшение размера LV

Проверка работы системы.

```
Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

This disk is currently in use - repartitioning is probably a bad idea.
It's recommended to umount all file systems, and swapoff all swap
partitions on this disk.


Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1):
First sector (2048-1049353, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-1049353, default 1049353): +100M

Created a new partition 1 of type 'Linux' and of size 100 MiB.
Partition #1 contains a LVM2_member signature.

Do you want to remove the signature? [Y]es/[N]o: y

The signature will be removed by a write command.

Command (m for help): w
The partition table has been altered.
Failed to add partition 1 to system: Device or resource busy

The kernel still uses the old partitions. The new table will be used at the next reboot.
Syncing disks.

root@localhost:/home/manieyvazi#
```

Рис. 9 — Уменьшение LV

Создание PV/VG/LV

-

Рис. 11 — Создание PV, VG и LV

Настройка fstab

•

```
GNU nano 8.1                                                    /etc/fstab

#
# /etc/fstab
# Created by anaconda on Fri Sep 26 13:01:01 2025
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=ecf62878-58a1-470b-a7cb-7da192353e46 /                       xfs     defaults        0 0
UUID=61940581-6552-45db-a269-f893ea8c6dac /boot                   xfs     defaults        0 0
UUID=09b04f01-5e64-4e8c-8f3c-b627431e957b none                    swap    defaults        0 0
/dev/vgdata/lvdata                        /mnt/data               ext4    defaults        1 2
/dev/vggroup/lvgroup                      /mnt/groups             xfs     defaults        1 2

#UUID=f989c7da-164e-42ad-b4b3-4687d8f43a65 /mnt/data              xfs     defaults        1 2
#UUID=cb353ca7-1f68-4b5a-bae6-64260c049ee3 /mnt/data-ext                  ext4    defaults        1 2
#UUID=63b3ad1f-e32c-47d1-bf0c-08a7ae8b8d38 none                   swap    defaults        1 2
```

Рис. 12 — Настройка fstab

Создание нового PV для расширения

•

```
Created a new partition 2 of type 'Linux' and of size 150 MiB.
Partition #2 contains a swap signature.

Do you want to remove the signature? [Y]es/[N]o: Y

The signature will be removed by a write command.

Command (m for help): t
Partition number (1,2, default 2): 2
Hex code or alias (type L to list all): 83

Changed type of partition 'Linux' to 'Linux'.

Command (m for help): t
Partition number (1,2, default 2): 2
Hex code or alias (type L to list all): 8e

Changed type of partition 'Linux' to 'Linux LVM'.

Command (m for help): p
Disk /dev/sdc: 1.5 GiB, 1610612736 bytes, 3145728 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x137c0a7c

Device     Boot   Start     End Sectors  Size Id Type
/dev/sdc1          2048 1230847 1228800  600M 8e Linux LVM
/dev/sdc2       1230848 1538047  307200  150M 8e Linux LVM

Filesystem/RAID signature on partition 2 will be wiped.

Command (m for help): w
The partition table has been altered.
Syncing disks.
```

Рис. 14 — Создание /dev/sdc2

Выводы по проделанной работе

Вывод

Были освоены процедуры работы с LVM: создание PV, VG, LV, монтирование, расширение и уменьшение томов. Отработаны навыки управления файловыми системами ext4 и XFS.